

Theoretical Analysis: Filter Performance for Salt & Pepper Noise

Local Filters - Noise Removal Analysis

Question 1: Which filter performed better according to PSNR?

Answer: The **Median Filter** achieved the highest PSNR (Peak Signal-to-Noise Ratio) value among all three filters.

Question 2: Does visual quality match PSNR values? Which filter preserves edges better?

Answer: Yes, the visual quality assessment correlates well with the quantitative PSNR measurements.

Visual Quality Comparison:

Filter	Visual Quality	Noise Removal	Edge Preservation
Median Filter	Excellent	Complete removal	Very good - edges sharp
Mean Filter	Poor - Blurred	Partial (but blurry)	Poor - edges smoothed
Gaussian Filter	Moderate	Some noise remains	Moderate

The visual quality matches PSNR values perfectly. The filter with highest PSNR (Median) also produces the best visual result and best edge preservation.

Question 3: Scientific reason for Median Filter superiority

Answer: The Median Filter is scientifically superior for salt & pepper noise because it is a **non-linear, rank-order filter** that is inherently **robust to outliers**.

Scientific Explanation:

1. Nature of Salt & Pepper Noise:

- Salt noise: Pixel value = 255 (white)
- Pepper noise: Pixel value = 0 (black)
- These are extreme outliers in the intensity range

- They do not represent true image information

2. Why Median Filter Works Best:

The median filter operates by:

- Examining a neighborhood around each pixel (e.g., $5 \times 5 = 25$ pixels)
- Sorting all pixel values in that neighborhood
- Replacing the center pixel with the **median value** (the middle value after sorting)

Result: Corrupted pixel values (0 and 255) are completely ignored!

3. Robustness Analysis for 5×5 Kernel:

- Total pixels in neighborhood: 25
- Maximum corrupted pixels median can handle: 12 (48%)
- This provides excellent noise tolerance

The Median Filter's superiority for salt & pepper noise removal stems from its fundamental mathematical property as a **rank-order statistic**. Unlike linear filters (Mean and Gaussian) that average values and are influenced by all pixels including outliers, the median filter completely rejects extreme outliers by selecting the central value of the sorted neighborhood. This property simultaneously achieves two critical goals:

1. **Effective noise removal** - Outliers are completely ignored
2. **Edge preservation** - The filter uses actual pixel values rather than creating artificial averages

This is why the Median Filter is the standard, go-to solution for salt & pepper noise removal in practical image processing applications.

Comparison of Filters for Salt & Pepper Noise Removal

Noise Statistics

